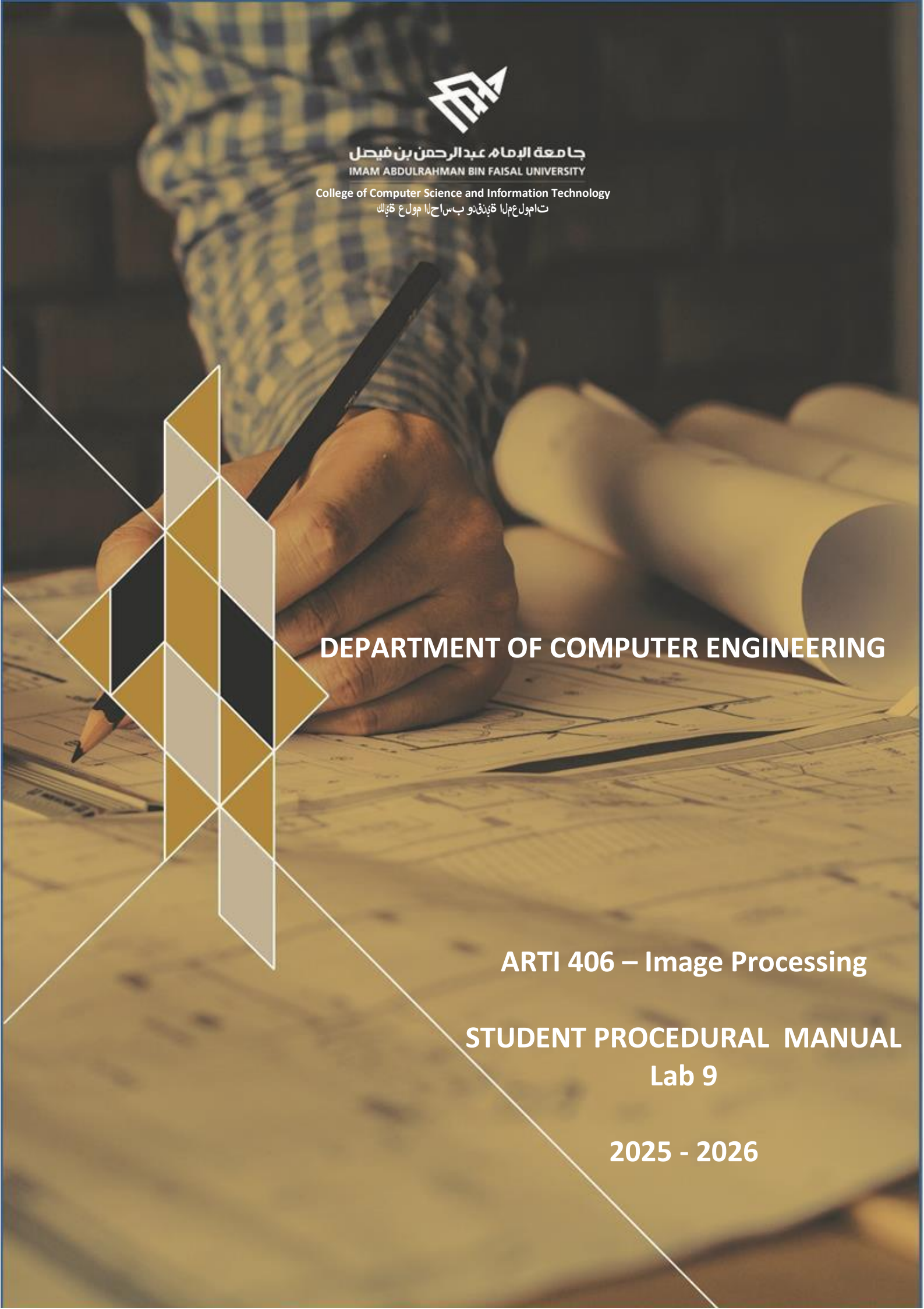


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DEPARTMENT OF COMPUTER ENGINEERING

ARTI 406 – Image Processing

STUDENT PROCEDURAL MANUAL
Lab 9

2025 - 2026

IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Lab 9 – Answers: Morphological Operations

Objective: Apply erosion, dilation, opening and closing to a new binary image, and see how the shape and size of the structuring element change the result.

Selected image: Balls.png (1274 × 1234 pixels), a black-and-white image of 15 numbered balls. It was thresholded with Otsu and inverted, so the balls are white (foreground) and the background is black.

Task 1: Apply Morphological Operations

All four operations use an 11 × 11 square kernel.

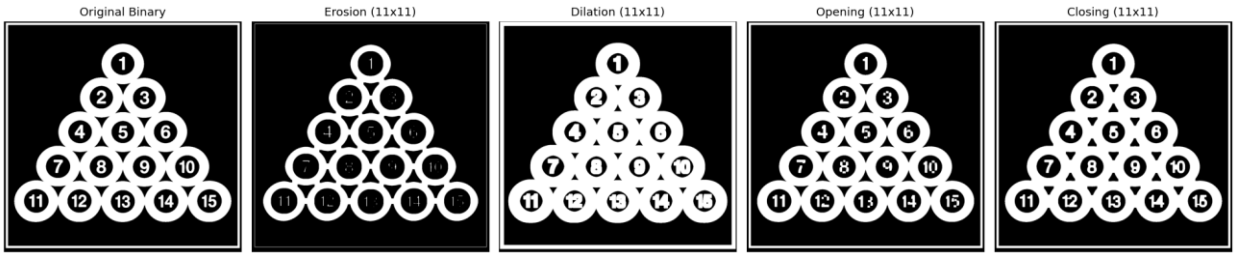


Figure 1: Original binary image and the results of erosion, dilation, opening and closing

Erosion: White objects shrink. The rings and digits get thinner, the thin frame almost disappears, and the black gaps between the balls get bigger.

Dilation: White objects grow. The rings, digits and frame get thicker, and the gaps between the balls become small triangles.

Opening: Erosion then dilation. It removes white details thinner than the kernel, so some digits break (e.g. 9 and 13), while the rings keep their size.

Closing: Dilation then erosion. It fills black gaps smaller than the kernel, so the gaps between the touching balls get smaller, while the rings keep their size.

Task 2: Experiment with Structuring Elements

Erosion and dilation were applied with square, cross and ellipse kernels of size 3 × 3, 5 × 5 and 7 × 7 (3 iterations, zoomed on balls 4, 5 and 6).

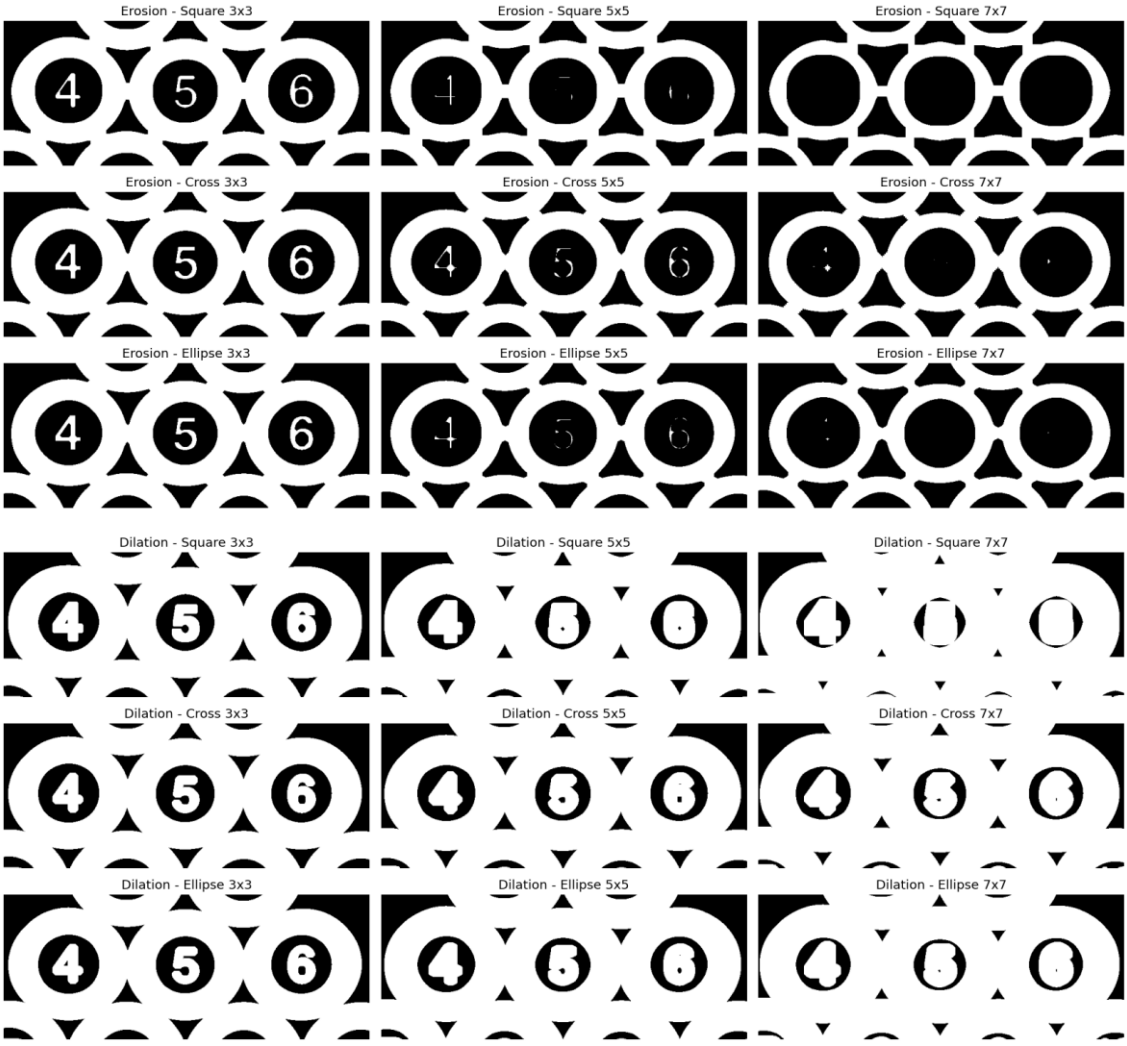


Figure 2: Erosion (top) and dilation (bottom) with different kernel shapes and sizes

Comparison

Size: A bigger kernel gives a stronger effect. With 3 × 3 the change is small; with 7 × 7 erosion removes the digits almost completely and dilation almost fills the gaps between the balls.

Square: Strongest effect, because it uses every pixel in the window (after 7 × 7 erosion only 48% of the white pixels are left). It also makes round edges look boxy.

Cross: Weakest effect, because it only uses the middle row and column (58% left after 7 × 7 erosion).

Ellipse: In between (55% left) and it keeps the round shape of the balls best, so it is the best choice for this image. Note: the 3 × 3 ellipse is the same as the 3 × 3 cross, so their results are identical.